Water Quality Analysis Report

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1. Dataset Overview

The cleaned dataset contains **3,276 water samples** across 9 chemical parameters. After cleaning, all missing values in pH (491), Sulfate (781), and Trihalomethanes (162) were filled using median imputation, ensuring no data loss.

Parameter	Min	Max	Mean	Std Dev
pH	1.93	13.21	7.14	1.44
Hardness (mg/L)	74.42	314.72	196.71	32.87
Solids (mg/L)	—	49,597	21,831	8,887

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2. Class Distribution

Out of 3,276 samples:

- **Not Potable (Unsafe): 1,966 samples → 60.0%**
- **Potable (Safe): 1,310 samples → 40.0%**

Inference: The dataset is **imbalanced** — 3 out of every 5 water samples are unsafe. This means any ML model trained on this data without handling imbalance will naturally be biased toward predicting "unsafe." This is why Logistic Regression accuracy stays around 60–62% — it's essentially learning the majority class.

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3. WHO Standard Violations

Parameter	WHO Limit	Samples Violating	% Violating
Solids	< 500 mg/L	3,256	**99.4%**
Sulfate	< 250 mg/L	3,225	**98.4%**
Chloramines	< 4 mg/L	3,204	**97.8%**
Conductivity	< 400 µS/cm	2,062	**62.9%**
pH	6.5 – 8.5	1,472	**44.9%**
Trihalomethanes	< 80 µg/L	635	**19.4%**
Turbidity	< 5 NTU	308	**9.4%**

Inference: Nearly the entire dataset violates WHO limits for Dissolved Solids, Sulfate, and Chloramines. This strongly suggests the data represents **heavily contaminated or industrial water sources**, not regular municipal water. Turbidity is the only parameter where most samples (90.6%) are within safe range.

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4. pH Analysis

- Overall mean pH: **7.14** — technically within neutral range
- But **44.9% of all samples** fall outside the WHO safe range of 6.5–8.5
- pH ranged from **1.93 (highly acidic)** to **13.21 (highly alkaline)** — extreme values indicating severe contamination in edge cases

pH Group Breakdown:

pH Group	Total Samples	Safe	Unsafe	% Safe
Acidic (< 6.5)	954	393	561	41.2%
Neutral (6.5–8.5)	1,804	695	1,109	38.5%
Alkaline (> 8.5)	518	222	296	42.9%

Inference: Surprisingly, pH group has almost **no impact on potability** in this dataset. Safe water percentages are nearly identical across Acidic (41.2%), Neutral (38.5%), and Alkaline (42.9%) groups. This tells us pH alone is not a reliable predictor of whether water is safe. . Water treatment cannot rely on pH correction alone.

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5. Chemical Comparison

Parameter	Safe Water Mean	Unsafe Water Mean	Difference
pH	7.146	7.132	+0.014
Hardness	196.71	196.71	~0
Solids	21,689	21,925	−236
Chloramines	7.115	7.106	+0.009
Turbidity	3.954	3.972	−0.018

Inference: The means of all 9 parameters are **almost identical** between safe and unsafe water. There is no single chemical parameter that clearly separates potable from non-potable water in this dataset. This is statistically confirmed by correlation analysis — the highest correlation any parameter has with Potability is just **−0.013 (Solids)**. This is a critical finding — it explains why simple ML models struggle on this dataset and why real-world water testing requires **multi-parameter combination analysis**, not individual thresholds.

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6. Correlation Analysis

All 9 parameters have near-zero correlation with Potability:

Parameter	Correlation with Potability
Solids	−0.013
Turbidity	−0.012
Organic Carbon	+0.010
Conductivity	+0.007
pH	+0.005
Chloramines	+0.003
Trihalomethanes	−0.002
Sulfate	~0.000
Hardness	~0.000

Inference: No linear relationship exists between any individual chemical parameter and water potability. This is a **non-linear classification problem** — meaning simple Logistic Regression will underperform, and this is expected and acceptable. A tree-based model like Random Forest would perform better.

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7. Outlier Analysis

All 9 parameters showed very low outlier counts (< 0.5% each), meaning the cleaned dataset is statistically well-behaved. However one important anomaly:

Solids column contains a negative value (−13,534 mg/L) — which is physically impossible since dissolved solids cannot be negative. This is a data quality issue in the original Kaggle dataset itself.

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8. Key Conclusions

1. Water scarcity of safe water is severe — Only 40% of sampled water is potable, meaning 60% fails safety standards. This reflects a real global water quality crisis.

2. No single parameter predicts potability — All correlations are below 0.02, meaning water safety is determined by the combined effect of multiple parameters, not any one chemical. This aligns with real ChE water treatment principles.

3. Dissolved Solids is the most widespread violation — 99.4% of samples exceed WHO limits for TDS. This indicates the dataset likely represents groundwater or industrial sources with high mineral content.

4. pH is not a reliable safety indicator — Safe and unsafe water show nearly identical pH distributions across Acidic, Neutral, and Alkaline groups. This challenges the common assumption that neutral pH = safe water.

5. Model limitation is dataset-driven — The ~62% Logistic Regression accuracy is not a coding failure — it reflects the genuine complexity of the problem. Even advanced models on this dataset typically cap at 67–70%, which you can cite from Kaggle discussions.

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9. Limitations

- Dataset source and collection methodology are not disclosed on Kaggle — geographic origin unknown
- Class imbalance (60:40) affects model training
- Missing values in pH, Sulfate, Trihalomethanes were imputed with median — introduces slight bias
- One negative Solids value detected — likely a data entry error in source data
- Logistic Regression assumes linear separability — not valid here, hence limited accuracy

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